

The Definition

If $E_1, E_2, \dots, E_n$ are **entity sets**, then a **relationship set** R is a **subset** of

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\} \setminus \{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

where $(e_1, e_2, \dots, e_n)$ is a **relationship instance**.

What does this mean?

1. Entity sets ($E_1, E_2, \dots, E_n$)

These are collections of entities of the same type.

- Example: $E_1 = \text{Students}$, $E_2 = \text{Courses}$

2. Cartesian product

The set

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

means **all possible combinations** of entities, one from each entity set.

- Example: (Ali, Database Systems), (Sara, Operating Systems), (Ali, Operating Systems), etc.

3. Relationship set (R)

A **relationship set** is **only a subset** of these possible combinations, not all of them.

- Why subset? Because not all entities are related — only specific valid pairs/triples/etc.
- Example: If Ali is enrolled in Database Systems but not in Operating Systems, then only (Ali, Database Systems) belongs to R .

4. Relationship instance

A single tuple (e.g., (Ali, Database Systems)) is called a **relationship instance**, showing a specific association between entities.

Example

- $E_1 = \{ \text{Ali}, \text{Sara} \}$ (Students)
- $E_2 = \{ \text{DB}, \text{OS} \}$ (Courses)

All possible combinations (Cartesian product):

$$\{(\text{Ali}, \text{DB}), (\text{Ali}, \text{OS}), (\text{Sara}, \text{DB}), (\text{Sara}, \text{OS})\}$$

Now suppose:

- Ali takes DB,
- Sara takes OS.

Then the **relationship set R (Enrolled-In)** is:

$R = \{(Ali, DB), (Sara, OS)\}$

Each pair here is a **relationship instance**.

Summary:

- The Cartesian product = all *possible* links.
- The relationship set = only the *actual* links that exist in reality.